Introduction to Data Science and Big Data

12th November 2024

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 Hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Marks

Question 1

a) Explain the key differences between data science, big data, **and** data analytics. **5**

b) Provide **ONE (1)** example where - data science, big data **and** data analytics overlap in real-world applications. **5**

Total 10 Marks

Question 2

a) Define the 5Vs of big data **and** explain why each is critical in managing large datasets. **5**

b) Identify data privacy and security issues in big data applications. For each issue explain how they are controlled. **5**

Total 10 Marks

Question 3

a) Define what is meant by the term **data**. **1**

b) Identify **and** explain **THREE (3)** primary categories of data. **6**

c) Provide **ONE (1)** real-world scenario example of each data type you identified in (b). **3**

Total 10 Marks

Question 4

a) Describe the concept of data preprocessing. Why is it important in the data science process? **5**

b) Explain the key steps involved in data preprocessing. **5**

Total 10 Marks

Question 5

a) Discuss the differences between descriptive and inferential statistics, including examples of each along with their purposes. **6**

b) Provide **ONE (1)** real-world example each of when descriptive **and** inferential statistics would be used in a data science project. **4**

Total 10 Marks

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Marks

Question 6

a) Explain the concept of Exploratory Data Analysis (EDA). **3**

b) Explain graphical **and** numerical methods used in EDA, with **TWO (2)** examples each and discuss **ONE (1)** difference between them. **7**

Total 10 Marks

Question 7

a) Explain in detail what is meant by ***feature engineering*** and discuss why it is important in machine learning. **5**

b) Give **ONE (1)** example of a feature engineering technique that improves model performance. Discuss the benefits of the technique you have identified. **5**

Total 10 Marks

Question 8

a) Discuss the main differences between supervised **and** unsupervised learning. **5**

b) Provide **ONE (1)** example of a problem where each type of learning is best suited. **5**

Total 10 Marks

Question 9

a) Discuss the role of Business Intelligence (BI) in modern organisations. **5**

b) Describe how BI tools can help in improving operational efficiency. Illustrate your answer with an example from any industry. **5**

Total 10 Marks

Question 10

a) Discuss the purpose and significance of ethics in data science. Illustrate your answer with examples. **5**

b) Explain FIVE (5) ways in which organisations ensure ethical use of data in their projects. **5**

Total 10 Marks

End of paper

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